A PRODUCT OF QUASI-COMPACT TOPOLOGICAL SPACES IS QUASI-COMPACT

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The subbasis theorem for quasi-compactness, through coverings by members of a subbasis. Invoked inside the counted proof (source line 2674). Reduces quasi-compactness of the product to finite refinability of coverings by subbasis members.

Source reference. Original lemma lemma-subbase-theorem: sources/stacks-topology/pinned-original-source.tex, source lines 2183–2183. Navigation pointer only: the counted text refers to the Stacks item lemma-subbase-theorem, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Theorem 0.1 (Tychonov)**.** *A product of quasi-compact spaces is quasi-compact.*

Proof. Let I be a set and for $i \in I$ let X_i be a quasi-compact topological space. Set $X = \prod X_i$. Let $\mathcal{B}$ be the set of subsets of X of the form $U_i \times \prod_{i' \in I, i' \neq i} X_{i'}$ where $U_i \subset X_i$ is open. By construction this family is a subbasis for the topology on X. By Lemma lemma-subbase-theorem it suffices to show that any covering $X = \bigcup_{j \in J} B_j$ by elements B_j of $\mathcal{B}$ has a finite refinement. We can decompose $J = \coprod J_i$ so that if $j \in J_i$, then $B_j = U_j \times \prod_{i' \neq i} X_{i'}$ with $U_j \subset X_i$ open. If $X_i = \bigcup_{j \in J_i} U_j$, then there is a finite refinement and we conclude that $X = \bigcup_{j \in J} B_j$ has a finite refinement. If this is not the case, then for every i we can choose an point $x_i \in X_i$ which is not in $\bigcup_{j \in J_i} U_j$. But then the point $x = (x_i)_{i \in I}$ is an element of X not contained in $\bigcup_{j \in J} B_j$, a contradiction. □

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